

DTSA 5800

Network Analysis for Marketing Analytics

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Elective	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective	Data Scientist (53.2 / 100)	41.5 / 100
SPECIALIZATION Marketing Analytics		

42

CURRENT AVG SCORE
across 6 job roles

81

POTENTIAL SCORE
with recommended additions

+65

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Network analysis is a long-standing methodology used to understand the relationships between words and actors in the broader networks in which they exist. This course will cover network analysis at it pertains to marketing data, specifically text datasets and social networks. Students will walk through a conceptual overview of network analysis, and dive into real-world datasets through instructor-led tutorials in Python. The course will conclude with a major project.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Data Scientist	53.2	96.1	+42.9	HIGH
Machine Learning Engineer	50.7	85.5	+34.8	HIGH
AI Engineer	44.6	78.0	+33.4	HIGH
ML Engineer	39.5	68.5	+29.0	MEDIUM
Applied Scientist	35.3	100.0	+64.7	HIGH
MLOps Engineer	25.8	60.5	+34.7	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 3 of 6 roles

systems

Business Communication 2 of 6 roles

insights

business

LLMs & Generative AI 2 of 6 roles

storage

Deep Learning 1 of 6 roles

training

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

- HIGH** **Add System Design coverage** — required vocabulary in 3 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.
- MEDIUM** **Consider Business Communication content** — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: insights, business.
- MEDIUM** **Consider LLMs & Generative AI content** — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: storage.
- LOW** **Optional: Deep Learning** — would broaden appeal to one additional role. Candidate terms: training.

GAP KEYWORDS BY ROLE

Data Scientist

53.2 → 96.1 (+42.9 pts)

analytics insights analytical build problems approaches

Machine Learning Engineer

50.7 → 85.5 (+34.8 pts)

learning analytics tools business mobile design science models

AI Engineer

44.6 → 78.0 (+33.4 pts)

models learning build model systems platform problems storage

ML Engineer

39.5 → 68.5 (+29.0 pts)

discovery learning machine systems models model world build

Applied Scientist

35.3 → 100.0 (+64.7 pts)

biology lab discovery scientific computational research storage problems

MLOps Engineer

25.8 → 60.5 (+34.7 pts)

model systems learning machine training tools models

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.